PYTHON ACTIVITY-BASED LEARNING

Case Study 02: Student Timetable Conflict Detection

Dr. RVR NRI Institute of Technology | Python Programming

Student Name: _____ **Roll No:** _____

Branch / Section: _____

Problem Statement

Compare class schedules represented by tuples and identify overlapping classes for the same student.

Objectives

• Represent timetable entries using tuples.

• Compare time intervals for clashes.

• Report conflicting course pairs.

Assumptions

• Time is represented in 24-hour HHMM integer format.

• A class overlaps another when its start is before the other's end and its end is after the other's start.

• All entries belong to the same student and same day.

Algorithm

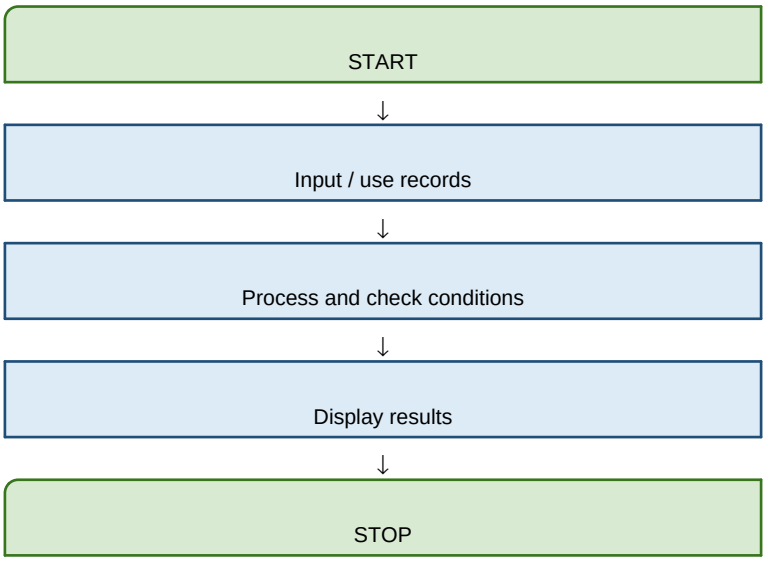
1. Create a list of course tuples containing course name, start time and end time.

2. Compare every pair using nested loops.

3. Check the interval-overlap condition.

4. Print each conflicting pair; otherwise print that no conflict exists.

Flowchart



Pseudocode

```
START
READ/store timetable tuples
SET conflict = False
FOR each pair of classes
    IF start1 < end2 AND start2 < end1
        DISPLAY the two courses
        SET conflict = True
IF conflict is False, DISPLAY no conflicts
STOP
```

Python Source Code

```
timetable = [
    ("Python", 900, 1000),
    ("Maths", 930, 1030),
    ("English", 1100, 1200),
    ("Physics", 1200, 1300)
]

conflict = False

for i in range(len(timetable)):
    for j in range(i + 1, len(timetable)):
        c1 = timetable[i]
        c2 = timetable[j]
        if c1[1] < c2[2] and c2[1] < c1[2]:
            print(c1[0], "conflicts with", c2[0])
            conflict = True

if not conflict:
    print("No timetable conflicts")
```

Test Cases

Test input / condition	Expected result
Python 09:00-10:00; Maths 09:30-10:30	Conflict: Python with Maths
English 11:00-12:00; Physics 12:00-13:00	No conflict: intervals touch only at boundary

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Python conflicts with Maths
```

Conclusion

The program uses tuples and nested loops to detect overlapping class periods.